



Unit No 5

UI Evaluation Techniques

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Unit -5 at a Glance



Unit-5: UI Evaluation Techniques

- What, why and when to evaluate
- Design guidelines
- Golden rules and heuristics
- Goals of Evaluation
- Evaluation criteria
- Evaluation through: Expert analysis
- User participation
- Testing techniques - Formative and Summative testing
- Surveys
- Peer reviews and so on



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What, why and when to evaluate



Introduction to UI Evaluation

Evaluating the user interface (UI) of a digital product is a crucial step in ensuring its success. By assessing the usability, aesthetics, and overall experience, designers and developers can identify areas for improvement and create more intuitive, engaging interfaces.

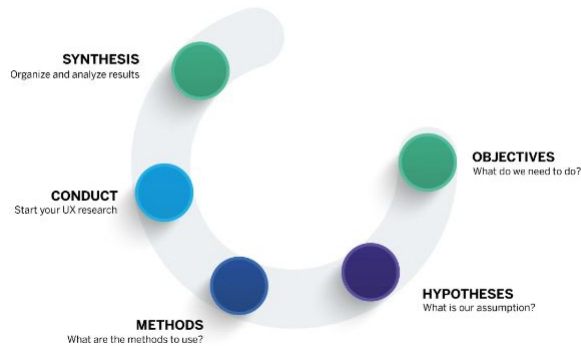


[dreamstime.com](https://www.dreamstime.com)

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Why Evaluate UI?

The UX Research Process



1

Improve Usability

UI evaluation helps identify pain points and areas where users struggle, allowing for targeted improvements.

2

Enhance User Experience

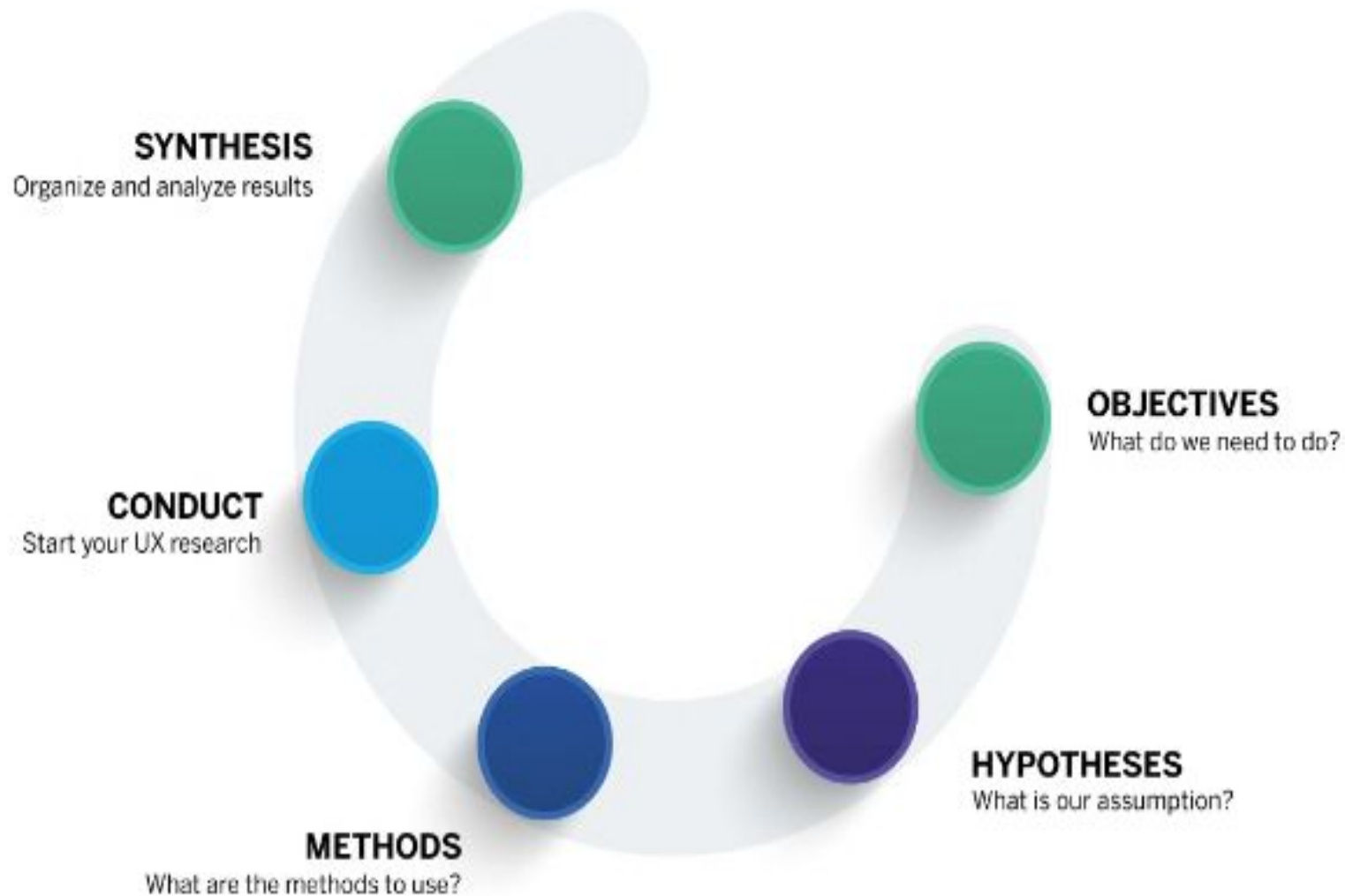
Evaluating the UI ensures the product is visually appealing, engaging, and easy to navigate.

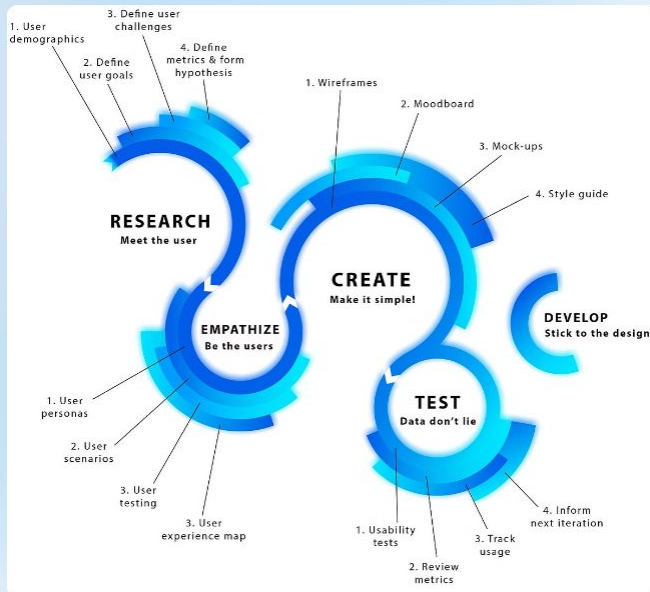
3

Increase Conversions

A well-designed UI can drive higher user engagement and lead to more successful outcomes, such as increased sales or sign-ups.

The UX Research Process





When to Evaluate UI

1

Ideation

Evaluate early wireframes and prototypes to validate design concepts and gather user feedback.

2

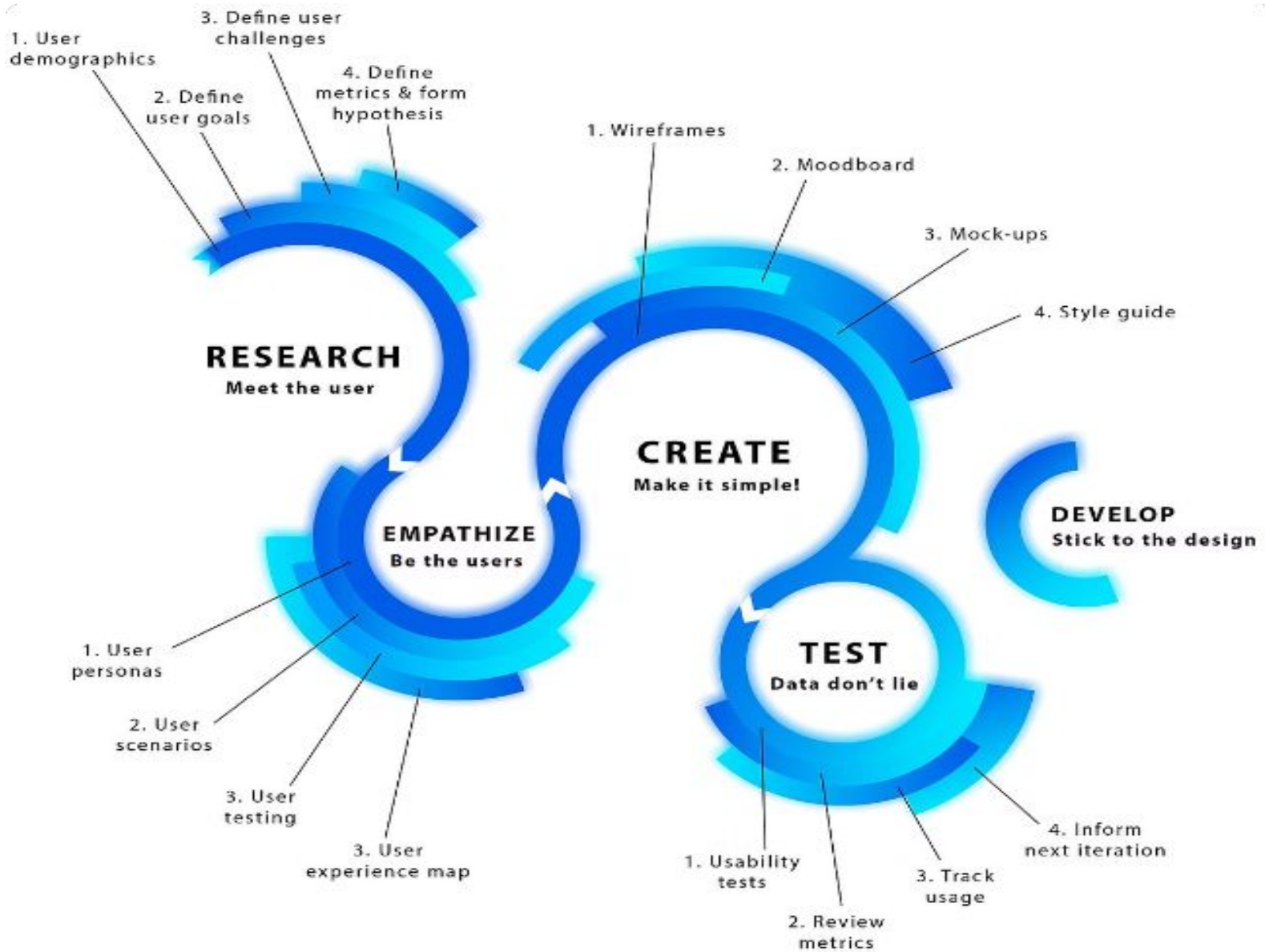
Development

Conduct usability testing throughout the development process to identify and address issues.

3

Launch

Continuously monitor user behavior and gather feedback to identify areas for ongoing improvement.



Types of UI Evaluation

Usability Testing

Observing users interacting with the UI to identify pain points and areas for improvement.

Heuristic Evaluation

Expert review of the UI based on established usability principles and best practices.

A/B Testing

Comparing user behavior between two or more UI variations to determine the most effective design.



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Design guidelines



Design guidelines

Design guidelines in UI (User Interface) are essential principles and best practices that help create effective, efficient, and enjoyable user experiences.

1. Consistency:

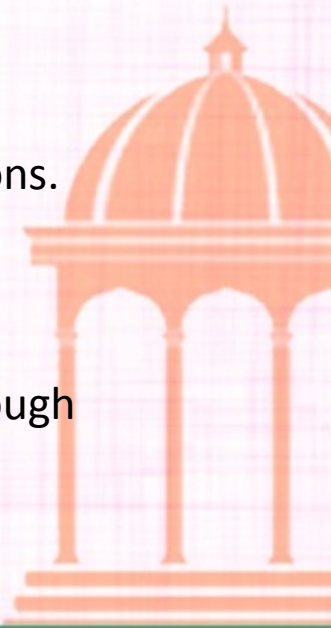
- **Visual Consistency:** Use uniform fonts, colors, and layouts.
- **Functional Consistency:** Ensure that similar actions lead to similar results.
- **Internal Consistency:** Maintain uniformity within the interface itself.
- **External Consistency:** Align with external interfaces that users are familiar with.

2. Simplicity:

- **Clarity:** Avoid unnecessary elements and focus on essential functions.
- **Minimalism:** Reduce clutter and present information concisely.
- **Task-focused:** Ensure the interface supports the user's tasks without distractions.

3. Feedback:

- **Immediate Feedback:** Provide instant responses to user actions.
- **Visibility of System Status:** Keep users informed about what is happening through appropriate feedback.



Design guidelines

4. Accessibility:

- **Universal Design:** Design for users with varying abilities.
- **Keyboard Navigation:** Ensure all functionalities are accessible via keyboard.
- **Contrast and Readability:** Use high-contrast color schemes and readable fonts for better accessibility

5. Error Prevention and Recovery:

- **Error Prevention:** Design to prevent errors before they happen.
- **Error Messages:** Provide clear, constructive error messages when errors occur.
- **Recovery:** Allow users to easily undo actions and recover from errors.

6. User Control and Freedom:

- **Flexibility:** Allow users to tailor the interface to their needs.
- **Undo and Redo:** Provide mechanisms to reverse actions.
- **Navigation:** Ensure easy navigation and freedom to explore without fear of getting lost.



Design guidelines

7. Affordance:

- **Interactive Elements:** Make it clear which elements are interactive.
- **Visual Cues:** Use visual indicators to suggest how elements can be used.

8. Performance:

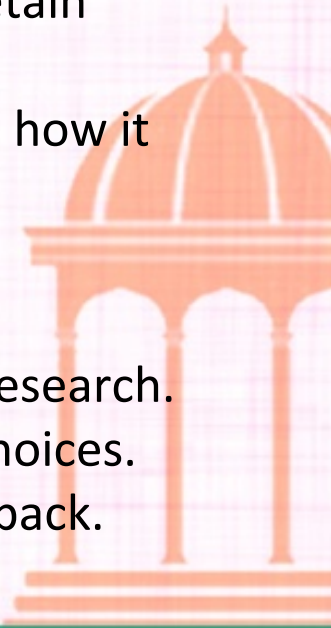
- **Efficiency:** Optimize for quick loading times and responsive interactions.
- **Scalability:** Ensure the interface performs well under different conditions and loads.

9. Aesthetics:

- **Appealing Design:** Create visually pleasing designs that attract and retain users.
- **Emotional Design:** Consider the emotional impact of your design and how it makes users feel.

10. User-Centered Design:

- **Research:** Understand user needs, behaviors, and contexts through research.
- **Testing:** Conduct usability testing with real users to validate design choices.
- **Iteration:** Continuously refine and improve the design based on feedback.





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Golden rules and heuristics



Shneiderman's Eight Golden Rules of Interface Design

Ben Shneiderman, a renowned computer scientist, proposed eight principles to guide the development of effective and user-friendly interfaces. These rules focus on creating intuitive, efficient, and satisfying experiences for software users.



Strive for Consistency

1

Consistent Patterns

Maintain consistent design elements, layouts, and terminology across the interface to help users feel comfortable and familiar with the system.

2

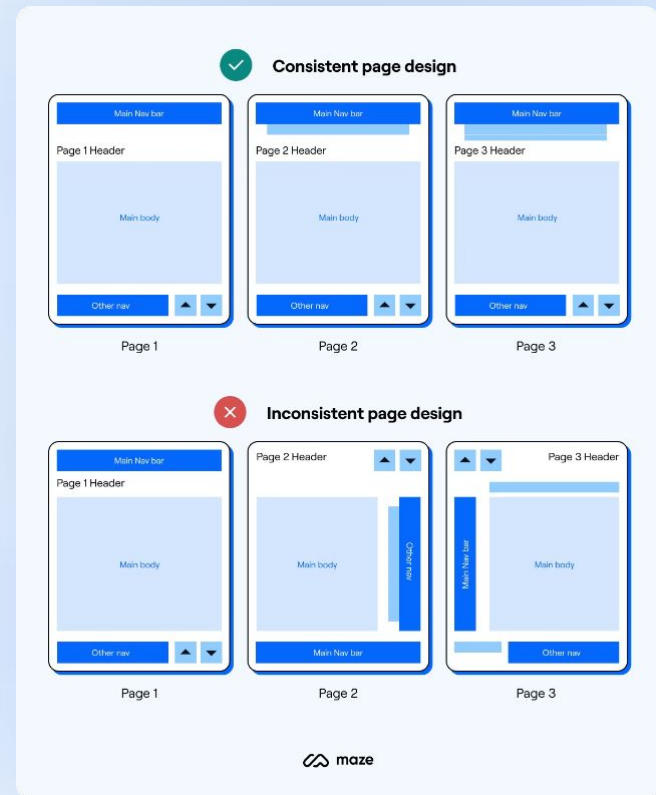
Predictable Behavior

Ensure that user actions produce predictable and expected responses, reducing confusion and frustration.

3

Familiar Conventions

Adhere to well-established design conventions and standards to leverage users' existing knowledge and expectations.



Enable Frequent Users to Use Shortcuts

Accelerate Productivity

Provide keyboard shortcuts, macros, and other time-saving features to help experienced users work more efficiently.

Personalize Experiences

Allow users to customize the interface and create their own shortcuts to fit their specific workflows and preferences.

Enhance Engagement

Empowering users with shortcuts fosters a sense of control and mastery, leading to increased user satisfaction and engagement.

Offer Informative Feedback

1

Provide Visibility

Keep users informed about the system's status and the progress of their actions through clear and timely feedback.

2

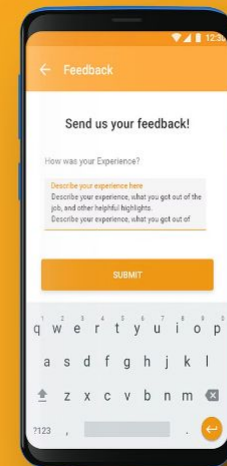
Reduce Uncertainty

Communicate the results of user actions and provide guidance to help users understand what is happening and what to do next.

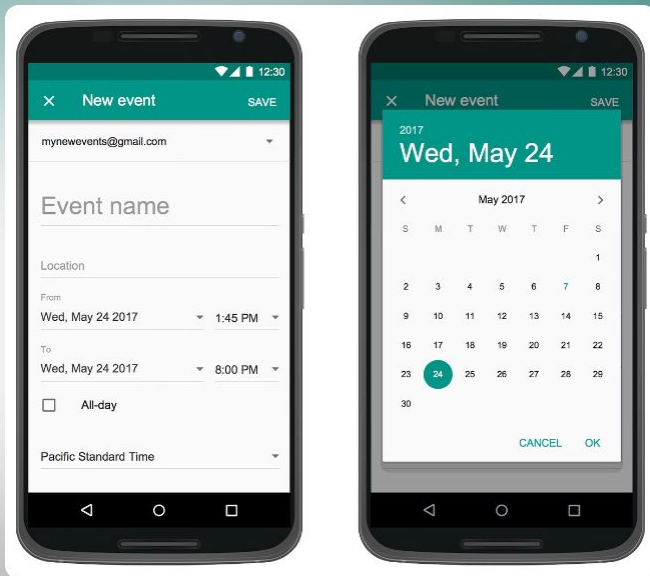
3

Build Trust

Responsive and informative feedback helps users feel in control and builds their confidence in the system.



Design Dialogs to Yield Closure



Clearly Defined Endpoints

Ensure that user interactions have a clear beginning and end, with a sense of completion upon task completion.

Avoiding Open-Ended Tasks

Provide users with a defined path forward and avoid leaving them wondering what to do next.

Satisfying Experiences

Designing dialogs with closure helps users feel a sense of accomplishment and satisfaction, leading to positive user experiences.

Offer Simple Error Handling



Gentle Guidance

Provide clear and helpful error messages that explain the problem and suggest possible solutions.



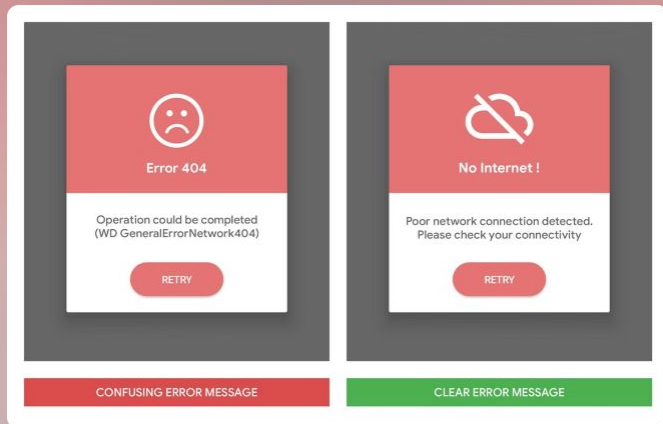
Easy Reversal

Allow users to easily undo or correct mistakes, minimizing the impact of errors.



Proactive Support

Anticipate and prevent common errors by designing the interface to be intuitive and forgiving.



Permit Easy Reversal of Actions

1

Minimize Anxiety

Allowing users to easily undo their actions reduces the fear of making mistakes and encourages exploration and experimentation.

2

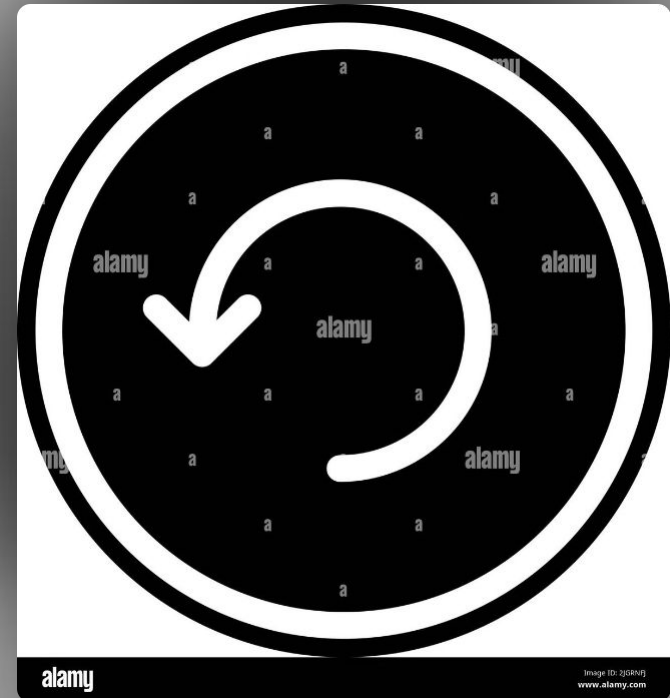
Facilitate Learning

The ability to reverse actions enables users to learn and improve their skills without the fear of permanent consequences.

3

Increase Confidence

Providing a safety net of easy reversal builds user confidence and fosters a positive, stress-free interaction experience.



Support Internal Locus of Control

Empower Users

Give users a sense of control and agency over the system, allowing them to initiate and direct actions.

Avoid Frustration

Preventing users from feeling like they are being manipulated or forced to take actions they don't want to take.

Foster Engagement

Enabling users to feel in charge of their experiences leads to increased motivation, satisfaction, and long-term engagement.

Reduce Short-Term Memory Load

1

Simplify Navigation

Organize the interface in a logical and intuitive way to minimize the cognitive effort required to find and use features.

2

Provide Visual Cues

Use visual elements like icons, labels, and grouping to help users quickly recognize and remember information.

3

Streamline Interactions

Design interactions to be as straightforward and effortless as possible, reducing the mental burden on users.

3 Minimize user's memory load



Slide adapted from Saul Greenberg



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Goals of Evaluation



Goals of Evaluation



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- Evaluation has three main goals
- 1. To assess the extent and accessibility of the system's functionality:** the design of the system should enable users to perform their intended tasks more easily. This includes not only making the appropriate functionality available within the system, but making it clearly reachable by the user in terms of the actions that the user needs to take to perform the task.
 - 2. To assess users' experience of the interaction:** This includes considering aspects such as how easy the system is to learn, its usability and the user's satisfaction with it. It may also include its enjoyment and emotional response, particularly in the case of systems that are aimed at leisure or entertainment
 - 3. To identify any specific problems with the system:** The final goal of evaluation is to identify specific problems with the design. This is, of course, related to both the functionality and usability of the design, it is specifically concerned with identifying trouble-spots which can then be rectified.





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Evaluation criteria



Evaluation criteria



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1. Effectiveness:

- How well does the interface enable users to achieve their goals accurately and completely?
- Are tasks performed correctly and with a low error rate?

2. Efficiency:

- How quickly can users complete tasks using the interface?
- Is the interface optimized for quick and smooth task completion?

3. Satisfaction:

- How enjoyable and pleasant is the interface to use?
- Are users satisfied with the overall experience?

4. Learnability:

- How easy is it for new users to learn to use the interface?
- Can users quickly become proficient with minimal training or guidance?

5. Memorability:

- How easily can users remember how to use the interface after a period of not using it?
- Does the interface design promote retention of key functions and features?



Evaluation criteria

6. Errors:

- How many errors do users make while using the interface?
- How severe are these errors, and how easily can users recover from them?

7. Accessibility:

- Is the interface usable by people with a wide range of abilities and disabilities?
- Does it comply with accessibility standards and guidelines?

8. Consistency:

- Is the interface consistent in its use of design elements and behaviors?
- Does it follow established conventions and standards?

9. Flexibility:

- Can the interface accommodate a range of user preferences and needs?
- Does it support customization and personalization options?

10. Feedback:

- Does the interface provide clear, timely, and useful feedback to users' actions?
- Is the system status always visible and understandable?

11. Aesthetic Design:

- Is the interface visually appealing and attractive?
- Does the design enhance the usability and user experience?





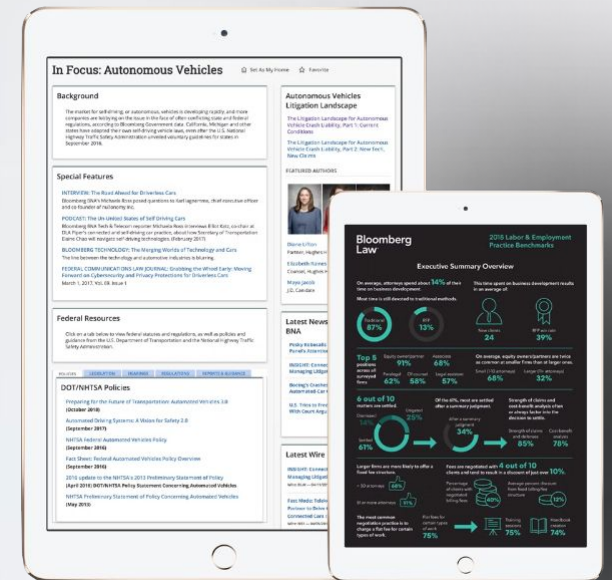
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Evaluation through: Expert analysis



Introduction to Expert Analysis

Expert analysis is a powerful tool for gaining deep insights and making informed decisions. By leveraging the specialized knowledge and experience of qualified professionals, organizations can uncover crucial insights to drive strategic success.



Defining Expert Analysis

1

In-Depth Assessment

Expert analysis involves a comprehensive examination of relevant data, information, and evidence to provide a thorough evaluation.

2

Specialized Expertise

The analysis is conducted by individuals with extensive knowledge, experience, and credibility in the subject matter.

3

Objective Insights

Experts provide unbiased, data-driven insights to help organizations make informed and strategic decisions.



Benefits of Expert Analysis

Enhanced Decision-Making

Expert analysis equips organizations with the information needed to make well-informed, strategic decisions.

Mitigated Risks

The insights provided can help identify and address potential risks, allowing for proactive risk management.

Competitive Edge

Leveraging expert analysis can give organizations a competitive advantage by providing unique and actionable insights.

Identifying Qualified Experts

Credentials and Certifications

Look for experts with relevant degrees, licenses, and certifications that demonstrate their expertise.

Extensive Experience

Seek out professionals with a proven track record of successful projects and in-depth industry knowledge.

Recognized Reputation

Consider experts who are respected authorities, published authors, or have a strong reputation in their field.

Professional References

Obtain references from past clients to verify the expert's competence, reliability, and quality of work.

SKILLS

- **Programming:** SQL, MATLAB, Java, Python (Pandas, scikit-learn)
- **Project Management:** Jira, Confluence, Agile, Kanban
- **Data Visualization:** Tableau, Metabase, Google Sheets, Salesforce
- **Certifications & Training:** CFA Level II (2021)

RESUME WORDED

Group skills by category

Simple lists save space

Skills must reflect the job description

Qualifications show technical skills

Gathering Relevant Information



1

Identify Data Sources

Gather all relevant data, documents, and information from internal and external sources.

2

Organize and Synthesize

Carefully structure and consolidate the information to facilitate the expert's analysis.

3

Verify Accuracy

Ensure the data is reliable, up-to-date, and comprehensive to support the expert's findings.

The framework is the coding plan to structure, label, and define data.

Sorting the data into a framework

Conducting the Analysis

1

Review Data

The expert thoroughly examines the gathered information to identify patterns, trends, and insights.

2

Apply Expertise

The expert leverages their specialized knowledge and experience to interpret the data and draw conclusions.

3

Validate Findings

The expert validates the analysis through additional research, testing, or expert consultations.

Interpretation of Findings

- Analyze both significant findings and not significant findings.
- Were the results what you predicted?
- Why do you think the results turned out the way they did?
- Were there any issues related to sampling, measurement, and procedural issues, as well as confounding variables?
- Provide possible explanations for the results.
- Link the results to any theoretical framework you used to develop your research question or hypotheses.

Interpreting the Findings



Critical Analysis

The expert examines the findings from multiple perspectives, considering potential implications and limitations.



Actionable Insights

The expert translates the analysis into practical, data-driven recommendations to guide decision-making.



Strategic Alignment

The expert ensures the findings are aligned with the organization's goals and strategic objectives.



Presenting the Evaluation

Clear Communication

The expert presents the analysis findings in a clear, concise, and accessible manner.

Compelling Narrative

The expert crafts a compelling narrative that connects the insights to the organization's needs.

Visual Aids

The expert utilizes effective visual aids, such as charts, graphs, and visualizations, to support the presentation.

Actionable Recommendations

The expert provides practical, data-driven recommendations that enable informed decision-making.



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Testing techniques - Formative and Summative testing



Formative Testing:

- Formative testing in UI (User Interface) design is a method used to evaluate and improve the usability and functionality of a design early in the development process.
- This type of testing is iterative and helps designers and developers identify issues, gather user feedback, and make necessary adjustments before the final implementation.



Formative Testing:



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key points about formative testing in UI:

- **Early Stage Evaluation:** Formative testing is conducted during the early stages of the design process. It allows designers to test concepts, wireframes, and prototypes.
- **User Involvement:** Real users are involved in the testing process to provide feedback. This ensures that the design meets the needs and expectations of the target audience.
- **Iterative Process:** Formative testing is iterative, meaning it is repeated several times throughout the design process. Each iteration helps refine and improve the design based on user feedback.



Formative Testing:



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key points about formative testing in UI:

- **Identifying Issues:** The primary goal is to identify usability issues, such as confusing navigation, unclear labels, or inefficient workflows, before they become more costly to fix later in the development process.
- **Feedback Collection:** Various methods can be used to collect user feedback, including usability testing sessions, interviews, surveys, and observation.
- **Improving Design:** The insights gained from formative testing are used to make informed design decisions and improvements. This leads to a more user-friendly and effective final product.



Formative Testing:



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key points about formative testing in UI:

- **Tools and Methods:** Common tools and methods used in formative testing include low-fidelity prototypes (like paper sketches), clickable wireframes, and interactive digital prototypes. User testing can be conducted in-person or remotely using screen sharing and recording tools.
- **Documentation:** Documenting the findings and changes made after each round of testing helps track progress and ensures that all feedback is addressed.



Summative Testing:

- Summative testing in UI (User Interface) design is a method used to evaluate the effectiveness and usability of a completed design.
- It is typically conducted at the end of the design process to ensure that the final product meets the intended goals and user requirements.



Summative Testing:

Key aspects of summative testing in UI:

- **Final Evaluation:** Summative testing is performed after the design and development phases are complete. It serves as a final check before the product is released to users.
- **Measurement of Success:** The primary goal is to measure how well the final design meets predefined usability criteria and user needs. This includes assessing task completion rates, error rates, and user satisfaction.
- **Quantitative and Qualitative Data:** Summative testing collects both quantitative data (e.g., time taken to complete tasks, number of errors) and qualitative data (e.g., user satisfaction, feedback on the overall experience).



Summative Testing:

Key aspects of summative testing in UI:

- **Controlled Environment:** Testing is usually conducted in a controlled environment to ensure consistency and reliability of the results. This can be done in usability labs or through remote testing tools.
- **Comparison to Benchmarks:** The results of summative testing are often compared to benchmarks or standards to determine the success of the design. This might include comparing against previous versions of the product or industry standards.
- **Formal Test Plan:** Summative testing typically follows a formal test plan that outlines the objectives, tasks, participants, and metrics to be used. This helps ensure a systematic and thorough evaluation.



Summative Testing:

Key aspects of summative testing in UI:

- **End-User Representation:** test participants should represent the target audience of the product. This ensures that the feedback and results are relevant and applicable to real-world usage.
- **Report and Recommendations:** The outcomes of summative testing are documented in a detailed report, highlighting strengths and weaknesses of the design.
- **Validation of Design Decisions:** Summative testing validates the design decisions made throughout the development process. It helps confirm that the design effectively solves the users' problems and delivers a positive user experience.
- **Compliance Testing:** In some cases, summative testing includes checking for compliance with accessibility standards, industry regulations, and other relevant guidelines to ensure the product meets all necessary criteria.





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Surveys



Surveys



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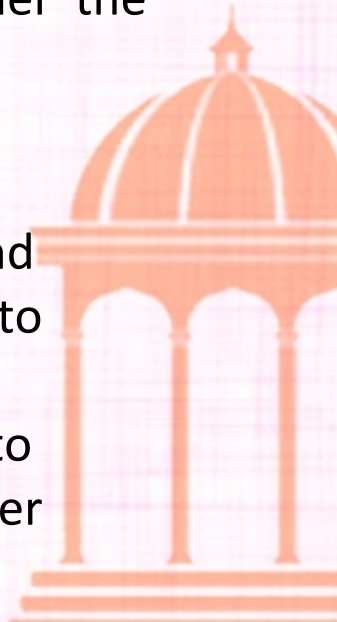
- Surveys are a valuable tool in UI (User Interface) evaluation to gather feedback from users about their experiences and perceptions.
- Surveys can provide quantitative and qualitative data that help identify areas for improvement and validate design decisions.

1. Define Objectives:

- **Purpose:** Clearly define the purpose of the survey. What specific information do you want to gather? Examples include assessing usability, understanding user satisfaction, or identifying pain points.
- **Target Audience:** Identify who will be taking the survey. Consider the users' demographics, experience levels, and familiarity with the UI.

2. Question Types:

- **Closed-Ended Questions:** Use multiple-choice, Likert scales, and rating scales to gather quantitative data. These questions are easy to analyze and can provide measurable insights.
- **Open-Ended Questions:** Include questions that allow users to provide detailed feedback in their own words. These can uncover insights that closed-ended questions might miss.



Surveys



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3. Survey Length:

- **Keep it Concise:** Ensure the survey is short and focused. Long surveys can lead to respondent fatigue and lower response rates.
- **Prioritize Questions:** Include only the most important questions that align with your objectives.

4. Question Clarity:

- **Clear and Simple:** Use straightforward language and avoid technical jargon. Ensure that questions are easy to understand.
- **Avoid Leading Questions:** Ensure that questions are neutral and do not bias the respondent towards a particular answer.

5. Logical Flow:

- **Organize by Topic:** Group related questions together to create a logical flow.
- **Progressive Disclosure:** Use conditional logic to show relevant questions based on previous answers.





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Peer reviews and so on



Peer reviews

- Peer review in UI (User Interface) evaluation involves colleagues or other experts assessing a design's usability, functionality, and overall effectiveness.
- This process helps identify strengths and weaknesses, leading to improvements before the design reaches end-users.

key steps of incorporating peer review into UI evaluation:

1. Preparation:

- **Define Goals:** Clearly outline what you want to achieve with the review, such as identifying usability issues, ensuring consistency, or improving visual design.
- **Gather Materials:** Collect all relevant design artifacts, such as wireframes, prototypes, user flows, and any existing user feedback.

2. Review Session:

- **Presentation:** Introduce the design, its objectives, target audience, and key decisions. Provide context so reviewers understand the goals and constraints.
- **Walkthrough:** Guide the reviewers through the design, explaining key screens, interactions, and user flows.
- **Feedback Collection:** Allow reviewers to explore the design independently and provide structured feedback. This can be done using a feedback form or checklist.

3. Feedback Documentation:

- **Record Comments:** Document all feedback received during the review session, categorizing it by type (e.g., usability, visual design, functionality).
- **Prioritize Issues:** Identify critical issues that need immediate attention and less urgent suggestions that can be addressed later

4. Follow-up:

- **Implement Changes:** Make the necessary adjustments based on the feedback. Prioritize changes that significantly impact user experience.
- **Iterative Review:** If significant changes are made, consider conducting another round of peer review to ensure the revisions are effective.
- **Communicate Updates:** Share the updated design with reviewers to show how their feedback has been incorporated.





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Exercise



Exercise



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1. What are the benefits and problems of using video in experimentation? If you have access to a video recorder, attempt to transcribe a piece of action and conversation. What problems did you encounter?
2. What are the factors governing the choice of an appropriate evaluation method for different interactive systems? Give brie





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THANK YOU

